

FRONTEND · UI · FLUTTER

Team Learning Roadmap

AI Interview Avatar — a 14-day plan to go from zero to demo-ready

Prepared for the Frontend / UI / Flutter squad · Timeline: 2 weeks max

What We're Building

An AI-driven avatar that interviews the user on camera and mic, then gives feedback on how they spoke and behaved. Your team owns everything the user sees and touches.



Flutter Web is the delivery surface for every one of these steps.

Why This Plan Is Different

A phase-by-phase Flutter course takes months. We don't have months — so this plan changes four things.



Parallel tracks, not phases

Each member specializes from Day 3 — nobody waits for everyone to finish the same phase.



Learn by building the real screens

Every Dart/Flutter lesson is applied directly to an actual app screen, not a throwaway tutorial.



Decisions locked on Day 1

One state manager, one avatar tech, one design tool. No mid-sprint tool debates.



3 days reserved for integration

The last stretch is protected for merging tracks, testing, and rehearsing the demo — not new features.

Team & Roles

Four tracks running side by side. Everyone learns the shared basics in Days 1–2, then specializes.



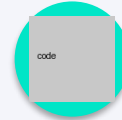
Leader — Frontend Lead

Architecture & code review
State management
API integration
Avatar integration



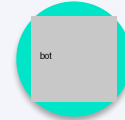
Member 1 — UI/UX

Figma design system
Responsive layouts
User flow
Reusable components



Member 2 — Flutter Dev

Dart & widgets
Screens & navigation
State wiring
Bug fixing



Member 3 — Avatar/Interaction

Avatar states & animation
Camera & mic handling
Cross-browser testing
Recording flow

DECISIONS

Tech Stack — Locked In

Committing to these now avoids losing days to tool debates mid-sprint.



Figma

Design & prototyping



Dart + Flutter Web

App language & framework



Riverpod

State management



Dio / http

REST API integration



Rive / Lottie

2D avatar animation



Camera & Mic Web APIs

Recording & permissions



Git — main/develop/feature

Branch strategy



Chrome, Edge, Safari

Test targets

14-Day Overview

WEEK 1 · Days 1–7

Learn + Build Foundations

Product, Dart & Figma basics together
Flutter fundamentals + first static screens
State machine wired, first avatar animation
Checkpoint: navigable app + permissions working

WEEK 2 · Days 8–14

Integrate + Polish + Rehearse

Real API wired into the interview flow
Avatar + camera/mic fully integrated
Cross-browser testing & bug fixing
Final rehearsal and demo-day backup plan

Week 1 — Foundations

Day	Focus	Owner(s)
Day 1	Product walkthrough, tech stack locked, GitHub repo + branch strategy set up	Whole team
Day 2	Dart essentials for devs; Figma basics & wireframe review for UI/UX; project skeleton created	Lead + all
Day 3	Flutter fundamentals (widgets, layout, navigation); design system finalized; avatar tech research	Flutter Dev / UI-UX / Avatar
Day 4	Build static screens from Figma; camera & mic permission flow; cross-browser setup	Flutter Dev + UI-UX / Avatar
Day 5	Riverpod wired to interview state machine; first avatar states (Idle, Talking); responsive pass	Lead + Flutter Dev / Avatar / UI-UX
Day 6	Checkpoint #1: internal demo — screens navigable, permissions working, avatar animating	Whole team
Day 7	Buffer / rest / optional catch-up	Whole team

Week 2 — Integration & Demo Prep

Day	Focus	Owner(s)
Day 8	REST/JSON basics; connect screens to a mocked backend response	Lead + Flutter Dev
Day 9	Real AI question/response flow wired; avatar reacts to state (Thinking, Listening)	Lead + Flutter Dev / Avatar
Day 10	Recording flow tied to interview state; error handling for denied permissions / network failures	Avatar + Flutter Dev
Day 11	Checkpoint #2: full flow assembled end-to-end, Landing to Report	Whole team
Day 12	Cross-browser & responsive testing (Chrome/Edge/Safari); bug bash	Whole team
Day 13	UI polish — animations, spacing, loading & error states	UI-UX + Flutter Dev
Day 14	Final rehearsal, demo script, recorded-video backup, submission	Whole team

Avatar Approach — Go With 2D

Option	Build Time	Web Performance	Fits 2 Weeks?
2D Avatar (Rive / Lottie)	Fast	Excellent	YES
3D Model + Lip Sync	Slow	Risky on Web	NO
Realistic AI Avatar (3rd-party)	Depends on vendor	Depends on vendor	RISKY



Recommendation

Build a 2D animated avatar in Rive (or Lottie) with 5–6 states: Idle, Listening, Thinking, Talking, Happy, Serious. It is fast to produce, performs reliably in the browser, and gives the AI something clear to trigger — 3D and realistic avatars are out of scope for a 2-week timeline.

Stay in Sync

Four tracks moving in parallel only works if everyone checks in daily.



Daily 15-min standup

What shipped yesterday, what’s today, and any blockers — same time every day.



End-of-day mini demo

Each track shows what got built that day, even if it’s rough.



Checkpoint #1 — Day 6

Foundations demo: navigable screens, permissions, first avatar animation.



Checkpoint #2 — Day 11

Full end-to-end flow demo, Landing through Report screen.



Lead reviews before merge

Nothing lands on develop without a quick review — catches issues early.

Risks We're Watching

Risk	Mitigation
Camera/mic behaves differently across browsers	Test on Chrome, Edge & Safari from Day 3–4, not Day 12
Avatar/AI response lag breaks the illusion	Avatar loops a "thinking" state, decoupled from real API latency
Backend/AI endpoint isn't ready in time	Build screens against mocked responses first, swap in real API later
Scope creep (3D avatar, extra features)	MVP scope locked on Day 1; extras go on a "later" list
New team members overwhelmed by Flutter	Pair programming + daily Lead reviews instead of solo learning

SUCCESS CRITERIA

Demo-Ready Checklist

By the end of Day 14, every item below should be true.



Full interview flow runs start to finish without crashing



Avatar shows at least 4 distinct states (idle, listening, thinking, talking)



App works cleanly on Chrome plus one more browser



Camera/mic permission denial doesn't break the app



Results / performance report screen displays after the interview



Team has rehearsed the live demo at least twice



Let's Build It

14 days. One MVP. One demo that works.

Questions and role assignments — let's lock them in today.